Bollinger Band Breakout Strategy

Technical System & Architectural Documentation (Pine Script v6)

System Overview

This algorithmic trading strategy identifies high-velocity volatility breakouts using Upper Bollinger Band penetrations validated by isolated institutional volume surges and multi-factor context confirmation. The system implements a robust automated state-tracking mechanism to avoid duplicate signals and features high-fidelity defensive loss-mitigation triggers to instantly protect capital from false breakouts.

Component	Mathematical Rule	Trading Purpose
Bollinger Bands	20 SMA baseline ± 2.0 Standard Deviations	Defines volatility boundaries & baseline trend.
Relative Strength Index	14-period bounded between 50 and 70	Ensures baseline momentum without exhaustion.
Volume Isolation	200% of previous bar > 2.0x macro background	Confirms true institutional participation.

1. Entry Conditions & Breakout Detection

A valid **BUY** signal requires four synchronous components to align on a single candle closure while the system state is flat (tradeState == 0):

- **Base Contained Structure:** The current candle open must sit completely inside the standard volatility limits.
- **Volatility Breakout:** The current candle close must cleanly clear the Upper Bollinger Band.
- **Relative Volume Spike:** Volume must double (2x) the immediate previous candle.
- **Relative Volume Isolation:** Historical volume from 1, 2, and 3 bars ago must all sit strictly below 50% of the current breakout candle's volume.
- **Goldilocks RSI Zone:** RSI must reside between 50 (momentum confirmed) and 70 (not overbought).

Technical Code Implementations:

```
baseVolumeSpike = volume >= (2 * prevVolume1)
halfOfBreakoutVol = 0.5 * volume
volumeIsIsolated = baseVolumeSpike and (prevVolume1 < halfOfBreakoutVol) and (prevVolume2 <
halfOfBreakoutVol) and (prevVolume3 < halfOfBreakoutVol)
```

```
breakoutTrigger = (open > bbLower and open < bbUpper) and (close > bbUpper) and volumeIsIsolated and
(rsiValue >= rsiMinLimit and rsiValue <= rsiMaxLimit)
```

2. Multi-Tier Exit Logic Hierarchy

When an active trade position is tracked (tradeState == 1), the script evaluates defensive and structural exits in strict hierarchical order:

Tier A: Bearish Volume Flood (Priority 1)

Monitors the first two candlesticks immediately following a trade entry. If a candlestick is bearish (red) and prints trading volume strictly greater than the initial breakout candle's massive entry volume, it detects an immediate institutional distribution block and closes the trade as a **VOL FLOOD** exit.

```
bearishVolumeFlood = barsSinceEntry <= 2 and (close < open) and (volume > entryVolume)
```

Tier B: Upper Band Re-Entry Loss Cut (Priority 2)

If the price closes back below the Upper Bollinger Band ceiling, it indicates an exhaustion of momentum. However, to allow a trade room to breathe on the immediate next candle, an exception rule is written: if candle 1 is bearish but its absolute body size is smaller than the breakout candle's body size, the exit is bypassed.

```
skipBandExitCondition = barsSinceEntry == 1 and (close < open) and (math.abs(close - open) <=
entryBodySize) and (close < bbUpper)
bandExitTrigger = useBandExit and (close < bbUpper) and not skipBandExitCondition
```

Tier C: Quick Reversal Fail-Safe (Priority 3)

If a candlestick is bearish and its volume exceeds 50% of the breakout entry candle's volume, it signals an immediate trend rejection, prompting a **FAIL EXIT**.

```
isFailExitPattern = (close < open) and (volume > 0.5 * entryVolume)
```

Tier D: Standard Volume Decay Exit (Priority 4)

If market activity dries up over 2 consecutive candle closes without hitting defensive stops, it exits via a normal trend expiration trend trailing rule.

```
isStandardExitPattern = (volume < prevVolume1) and (prevVolume1 < prevVolume2)
```

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